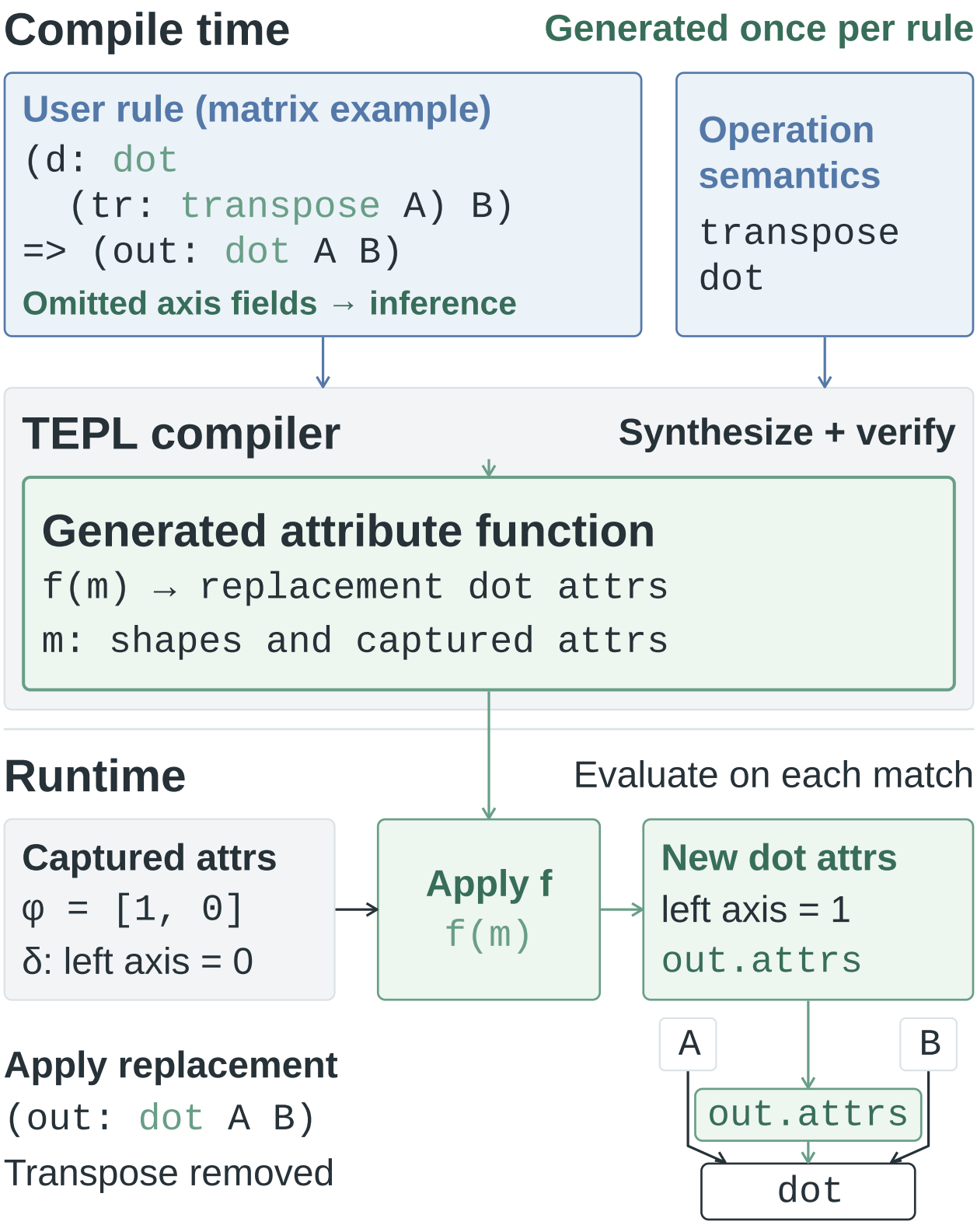

Compile time
Generated once per rule
User rule (matrix example)
(d: dot
(tr: transpose A) B)
=> (out: dot A B)
Omitted axis fields → inference
Operation semantics
transpose
dot
TEPL compiler
Synthesize + verify
Generated attribute function
f(m) → replacement dot attrs
m: shapes and captured attrs
Runtime
Evaluate on each match
Captured attrs
φ = [1, 0]
δ: left axis = 0
Apply f
f(m)
New dot attrs
left axis = 1
out.attrs
Apply replacement
(out: dot A B)
Transpose removed
A
B
out.attrs
dot
Only the left contraction axis is shown.